SUBSCRIPTION CHURN AND RETENTION RADAR

Project Progress Summary & Final Checklist

Prepared from the project brief and the SQL steps discussed in the main and branch chats.

Business question
Which customer segments churn, when do they leave, and how much revenue is at risk?

Database: churn_project
SQL environment: MySQL Workbench
Dataset: synthetic/sample data

Note: this summary records the work discussed and results explicitly reported. It does not claim that every query output was independently checked.

1. Project Overview

The project uses SQL to explore subscription churn, retention patterns, customer value, and possible churn risk. The aim is to turn query results into useful actions for a customer-retention team.

Churn definition currently used: a customer is marked churned when the subscription/churn flag indicates cancellation. This is a project definition applied to synthetic data, not a validated company policy.

Objectives

• Create a customer-centred relational schema.
• Calculate overall and plan-wise churn.
• Estimate monthly revenue at risk.
• Examine payment failures and low usage.
• Build cohort retention and RFM views.
• Compare monthly usage with LAG()/LEAD().
• Assign risk bands and recommend actions.
• Prepare an executive summary and final report.

2. Database Tables

Table	Purpose	Progress
customers	Customer master: ID, name, signup date, plan, monthly fee.	Created; data insertion reported.
subscriptions	Subscription dates and status.	Created; data insertion reported.
payments	Payment date, amount, status.	Created; data insertion reported.
usage_events	Sessions and minutes used.	Created; data insertion reported.
support_tickets	Issue category and ticket status.	Table appears in SHOW TABLES.
churn_flags	Churn indicator used by analysis queries.	Table appears in SHOW TABLES.
monthly_usage_history	Monthly activity used for cohort and LAG/LEAD.	Created and populated after a missing-table error; user reported verification complete.

3. Analysis Queries Discussed

Queries 1–10 below were provided across the chats. The user reported running the queries, but most individual result grids were not shared. Only the explicitly reported numeric result is presented as a known result.

#	Analysis	Status / purpose
1	Total churn rate	Query provided and reported run; calculates total, churned, retained and churn percentage.
2	Plan-wise churn	Compares churn by plan type.
3	Revenue at risk	Reported output: 100 churned customers and ■36,500 monthly revenue at risk.
4	Payment failure analysis	Compares payment status and churn.
5	Low usage analysis	Examines customers with very low usage.
6	Risk-level count	Groups customers into Already Churned, High, Medium and Low Risk.
7	Monthly cohort retention	Missing-table error addressed by creating/populating monthly_usage_history; user said Step 1 was complete.
8	RFM segmentation	Basic recency/frequency/monetary query and a simple spending-based segment query provided.
9	LAG()/LEAD()	Compares each customer's usage with previous and next monthly records; user reported verification complete.
10	Risk score + recommendations	Sample rule-based risk score and suggested action per customer provided; outputs should be checked.

4. Dataset and Interpretation Notes

The sample data is synthetic, not real company data. Earlier steps described around 500 records per main table; exact row counts should be checked in the current database if needed for submission.

Important limitations

• Monthly usage history was generated for up to six months from each customer's signup month, so the retention timeline is limited.
• The basic payments data has one payment per customer. Frequency may therefore equal 1 for everyone and is not a meaningful long-term frequency measure.
• The risk score is a hand-built illustrative rule, not a trained machine-learning prediction model.
• The cohort query treats minutes_used > 0 as activity. Since the usage data is generated, retention may appear artificially high.
• Support-ticket joins can duplicate customers if a customer has multiple tickets.

5. Known Result

Reported result: 100 churned customers and ■36,500 monthly revenue at risk.

Do not invent the overall churn percentage, plan-wise churn rates, retention percentages, RFM segment counts, or risk-level counts. Copy those numbers from the actual MySQL result grids.

6. Executive Summary Draft

This project uses MySQL to analyse subscription churn and retention using a synthetic dataset. The analysis covers overall churn, plan-wise churn, revenue at risk, payment failures, low usage, cohort retention, RFM-style segmentation, monthly usage comparisons with LAG()/LEAD(), and a rule-based churn-risk score. The reported result identifies 100 churned customers and ■36,500 in monthly revenue at risk. Suggested actions include resolving failed payments, engaging low-usage customers, closing open support tickets, and running win-back campaigns for customers already marked as churned. Because the dataset is synthetic and the risk score is rule-based, these findings demonstrate SQL analytics rather than a validated prediction model.

7. Final Report Structure

1. Title page
2. Problem statement and objectives
3. Churn definition and database schema
4. Dataset description and limitations
5. SQL analysis queries and output tables
6. Cohort retention and RFM analysis
7. LAG()/LEAD() comparison
8. Risk score, risk bands and recommended actions
9. Executive summary
10. Conclusion

8. Final Completion Checklist

Task	Status from chat	Next action
Tables and data	Reported created/inserted	Keep SQL scripts and confirm row counts if required.
Queries 1–6	Provided and reported run	Use actual result grids in the report.
Cohort retention	Table issue addressed; user reported check complete	Save output and mention synthetic-history limits.
RFM	Basic query + simple segment query provided	Mention the one-payment-per-customer limitation.
LAG()/LEAD()	Query provided; user said Step 1 complete	Save result output/screenshot.
Risk score	Query provided	Confirm output and include recommended action.
Executive summary	Draft included above	Update with actual query results.
Final submission report	This is a progress-summary PDF, not a results-complete final submission report	Add real result tables/screenshots and college details if needed.

This PDF summarises the project work and next steps. To make the final college-submission report complete, add the actual SQL outputs/screenshots and avoid presenting synthetic results as real-world findings.